25901045 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular EXAMINATION November 2025

First SEMESTER

Skill Development Components (SDC) - MG1SDCIIM102 - MATHEMATICS FOR INSTRUMENTATION

(2025 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer any **5** questions

Each question carries **2** marks

1. Define a vector quantity. Give an example. [A] / [CO1]
2. Represent $|\vec{a} \times \vec{b}|$ geometrically [A] / [CO1]
3. Find the adjoint of the matrix A if the cofactor of the matrix A is $\begin{bmatrix} 9 & 8 & 7 \\ 3 & 4 & 1 \\ 6 & 0 & 5 \end{bmatrix}$ [A] / [CO2]
4. Define a Column matrix. Give an example. [A] / [CO2]
5. If $\frac{dy}{dx} = x^3$, find y [An] / [CO3]
6. If y=f(x), then define the derivative of y with respect to x [An] / [CO3]
7. Write the Zero property in Boolean Algebra [An] / [CO4]
8. Find the value of the Boolean expression $1.\overline{1} + (\overline{0.1} + 0.1).\overline{1}$ [An] / [CO4]

[2x5 = 10]

Part B

Short Answer Questions

Answer any **5** questions

Each question carries **6** marks

9. Find the Dot product and Cross product of the vectors $\vec{a} = \hat{i} + 2\hat{j} - 4\hat{k}$, $\vec{b} = 5\hat{i} - 3\hat{j} + 2\hat{k}$. Comment whether they are parallel or perpendicular [A] / [CO1]

10. If $\vec{a} = \hat{i} + 2\hat{j} - \hat{k}$, $\vec{b} = \hat{i} - 3\hat{j} + 2\hat{k}$, $\vec{c} = 2\hat{i} - 3\hat{j} + \hat{k}$, find $(\vec{a} + 2\vec{b}) - (\vec{c} + \vec{b})$ [A] / [CO1]

11. If A= $\begin{bmatrix} 1 & 3 & 7 \\ 4 & 4 & 3 \\ 2 & 1 & 2 \end{bmatrix}$, Prove that B is Symmetric if B =½[A + A^T] [A] / [CO2]

12. Define idempotent matix. Check whether A= $\begin{bmatrix} 1 & 3 & 1 \\ 0 & 4 & 1 \\ 2 & 1 & 2 \end{bmatrix}$ is idempotent or not [A] / [CO2]

13. Find $\int (x^{-5} + 2x^3 - 5x^2 + 2x)dx$ [An] / [CO3]

14. Identify at which point the graph $y = 3x^2 - 5x + 1$ has an horizondal tangent. [An] / [CO3]

15. If F = $xz + yz$ and G = $\overline{xyz} + xyz$, find the values F+G and F.G using tables. [An] / [CO4]

16. Find the sum of product expansions for the function $F(x, y, z) = \overline{x} + y + \overline{z}$ [An] / [CO4]

[6x5 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **15** marks

17. Find the area of the triangle with vertices (1,0,3), (2,-1,2), (1,2,5). Check whether it is a right triangle. [An] / [CO1]

18. Solve: 4x -3y = 1; 2x + y = 3 using i) Cramers rule ii) Matrix inversion method [A] / [CO2]

19. If $u = (x^2 + 3)$ and $v = (x + 4x^2)$ find i) $\frac{d(uv)}{dx}$ using product rule ii) $\frac{d(\frac{u}{v})}{dx}$ using quotient rule [An] / [CO3]

20. i) Explain AND gate [An] / [CO4]
ii) Construct a circuit having the out put $(x + y + z)(\overline{x} + \overline{y} + \overline{z})$
iii)Construct a circuit with three varibles having only OR and Inverter gate.Find the out put

[15x2 = 30]